Human Activity Recognition in Older Adults Using Dual Accelerometer Data and Machine Learning

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Abstract. This paper presents a machine learning pipeline for human activity recognition (HAR) targeting older adults, a population largely underserved by existing models trained on young, healthy individuals. Using the HAR70+ dataset — dual accelerometer recordings from 18 participants aged 70–84, including walking-aid users — we classify seven daily activities: walking, shuffling, stair climbing up, stair climbing down, standing, sitting, and lying. Raw signals from lower back and thigh sensors are preprocessed via per-subject linear interpolation and Savitzky–Golay filtering, then segmented using a 100-sample (2.0 s) sliding window with 50% overlap. Each window is encoded as a 56-dimensional feature vector of time-domain and frequency-domain statistics across eight signal channels. Six classifiers are evaluated — Decision Tree, SVM, Naive Bayes, Random Forest, AdaBoost, and XGBoost — with macro-averaged F1 as the primary metric to account for heavy class imbalance. XGBoost achieves the best standard-split performance (macro F1 = 0.8193, weighted F1 = 0.9618), while Leave-One-Subject-Out (LOSO) evaluation reveals a substantial generalization gap, with Random Forest dropping from F1 = 0.9420 to 0.6696 across unseen individuals. Static activities such as sitting and lying are classified near-perfectly, while shuffling and stair-related classes remain the most challenging due to biomechanical similarity and severe underrepresentation. A 3D sequence tensor is additionally prepared for future CNN and LSTM experimentation. These results confirm the value of gradient-boosted ensembles for structured HAR feature data while highlighting subject-independent evaluation as essential for realistic performance estimation in health monitoring applications targeting older populations.

1 Introduction

1.1 Background

Human activity recognition (HAR) is an active area of research with applications in healthcare monitoring, fall detection, and assistive technology. The majority of existing HAR systems are trained on data collected from young, healthy adults, which limits their ability to generalize to older populations whose movement patterns differ significantly due to age-related changes in gait, balance, and mobility.

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1.2 Motivation

This work addresses that gap by developing a machine learning pipeline for activity classification using the HAR70+ dataset, which contains dual accelerometer recordings from adults aged 70 and above, including individuals who use walking aids. The dataset captures seven daily activities — walking, shuffling, stair climbing up, stair climbing down, standing, sitting, and lying — recorded at 50 Hz using sensors placed on the lower back and thigh.

1.3 Project Pipeline

The pipeline covers the full recognition workflow: signal preprocessing, windowing, feature extraction, and classification using both classical machine learning models and deep learning approaches. A key focus of this work is evaluating model performance under both standard train-test splits and subject-independent evaluation schemes such as Leave-One-Subject-Out (LOSO), which better reflects real-world deployment conditions where the model must generalize to unseen individuals.

2 Dataset Exploration

2.1 Dataset Overview

The HAR70+ dataset consists of recordings from 18 older adult participants, with subject IDs ranging from 501 to 518. Each participant completed a single recording session, yielding one day of data per subject. The dataset contains a total of 2,259,597 samples across all subjects, recorded at a sampling frequency of 50 Hz using two inertial measurement units placed on the lower back and right thigh. Each sensor records acceleration along three orthogonal axes (x, y, z), resulting in six raw signal channels: `back_x`, `back_y`, `back_z`, `thigh_x`, `thigh_y`, and `thigh_z`.

2.2 Participant Demographics

The participant pool spans ages 70 to 84, with a mean age of approximately 76 years. The cohort includes 10 female and 8 male participants, providing moderate demographic diversity within the older adult age group. Table 1 summarizes the per-subject statistics including gender, age, number of recorded samples, and recording duration.

Table 1: HAR70+ Dataset — Per-Subject Summary Table. 18 participants (ages 70–84), including 10 female and 8 male subjects, across 7 activity classes.

Subject ID	Gender	Age	Samples	Unique Labels	Start Time	End Time	Duration (s)
501	F	74	103860	7	14:42:03	15:17:06	2102.2
502	F	71	131367	6	08:23:07	09:07:50	2682.7
503	F	75	116413	5	14:42:07	15:23:46	2498.6
504	M	73	150758	7	10:13:54	11:05:56	3121.4
505	M	76	87006	5	14:42:30	15:11:49	1758.5
506	F	77	122714	7	14:28:07	15:11:41	2614.2
507	M	78	120125	7	14:27:16	15:10:05	2569.0
508	F	74	130494	7	14:30:21	15:15:06	2685.4
509	M	79	121763	7	14:07:24	14:49:25	2521.8
510	F	80	122061	7	08:17:05	08:58:45	2499.6
511	M	71	128063	7	10:10:25	10:54:24	2638.7
512	F	82	119310	7	15:39:36	16:21:55	2538.5
513	M	83	123599	5	11:11:42	11:53:52	2530.6
514	F	72	101510	5	14:13:04	14:47:41	2076.7
515	M	84	153517	5	09:28:14	10:20:12	3118.4
516	F	70	138278	7	13:34:08	14:22:14	2886.6
517	M	76	147045	7	10:04:08	10:54:18	3009.6
518	F	73	141714	6	10:24:37	11:13:14	2916.2

2.3 Class Distribution

Seven activity classes are represented in the dataset: walking (1), shuffling (3), stairs up (4), stairs down (5), standing (6), sitting (7), and lying (8). Note that label 2 is absent from the data. The class distribution is highly imbalanced, with walking accounting for approximately 1.08 million samples — nearly half the entire dataset — while stairs up and stairs down together represent fewer than 20,000 samples. Standing and sitting are moderately represented at approximately 420,000 and 480,000 samples respectively, while lying accounts for around 200,000 samples and shuffling for approximately 63,000 samples. This imbalance is an important consideration during model training and evaluation, as shown in Figure 2.

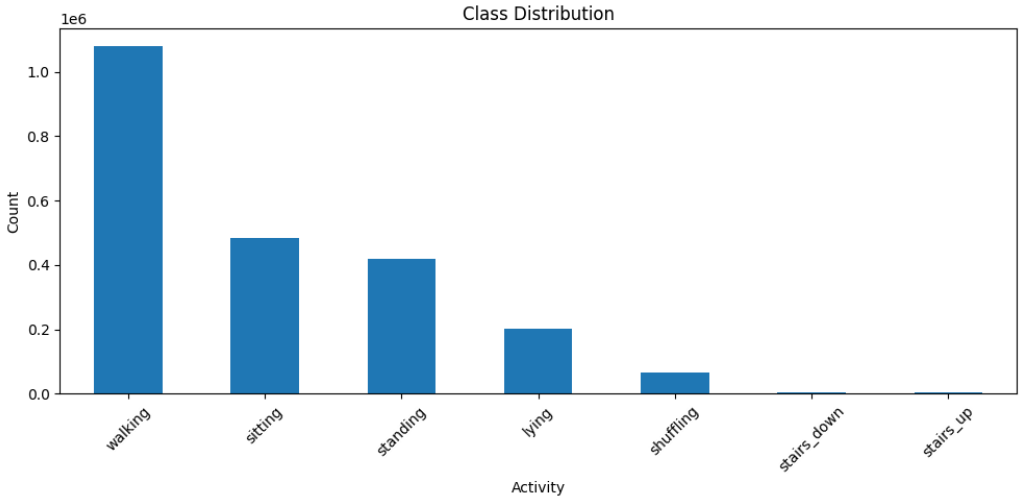


Figure 1: Class distribution across all 18 subjects. Walking is the dominant class, while stair-related activities are significantly underrepresented.

3 Annotation and Signal Exploration

3.1 Walking

During walking, both the back and thigh sensors exhibit clear periodic oscillations corresponding to individual footsteps. The `thigh_x` channel shows the most pronounced variation, with repeated sharp spikes ranging between approximately -1.0 and -2.5 g, reflecting the impact loading at each heel strike. The back sensor axes show moderate oscillation around zero, capturing the rhythmic torso sway that accompanies normal gait, as shown in Figure 3.



Figure 2: Class distribution across all 18 subjects. Walking is the dominant class, while stair-related activities are significantly underrepresented.

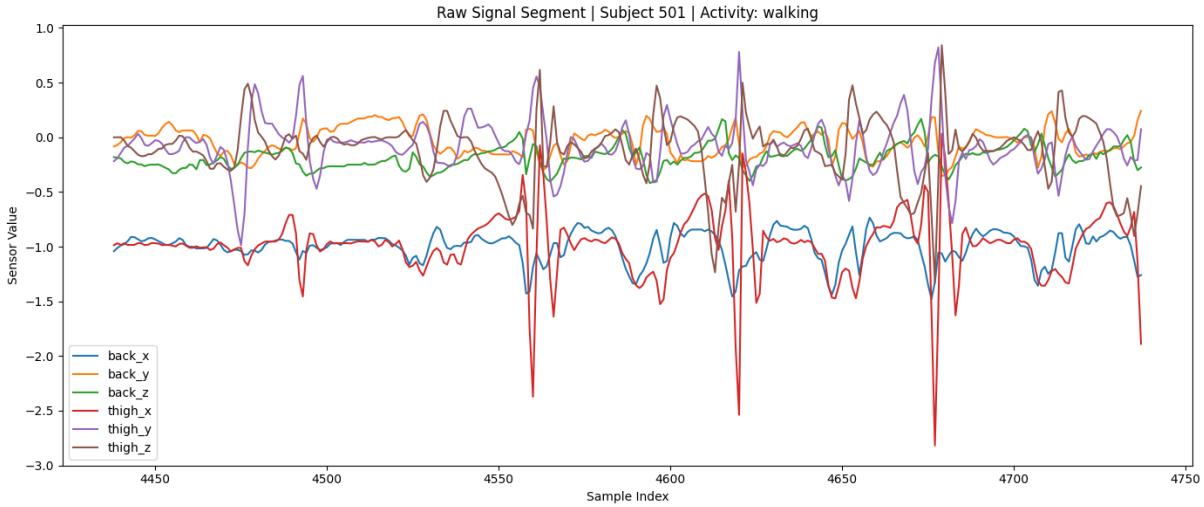


Figure 3: Raw accelerometer signal during walking — Subject 501. Periodic spikes in `thigh_x` reflect individual footsteps at approximately 50 samples apart.

3.2 Sitting

During sitting, all six channels produce very low-variance, near-constant signals consistent with a static posture. The `back_z` channel stabilizes at approximately $+0.3$ g and `thigh_z` at approximately $+0.1$ g, both reflecting gravitational alignment in a seated position. The `back_x` and `thigh_x` channels remain near -1.0 g with minimal fluctuation throughout the segment, as shown in Figure 4.

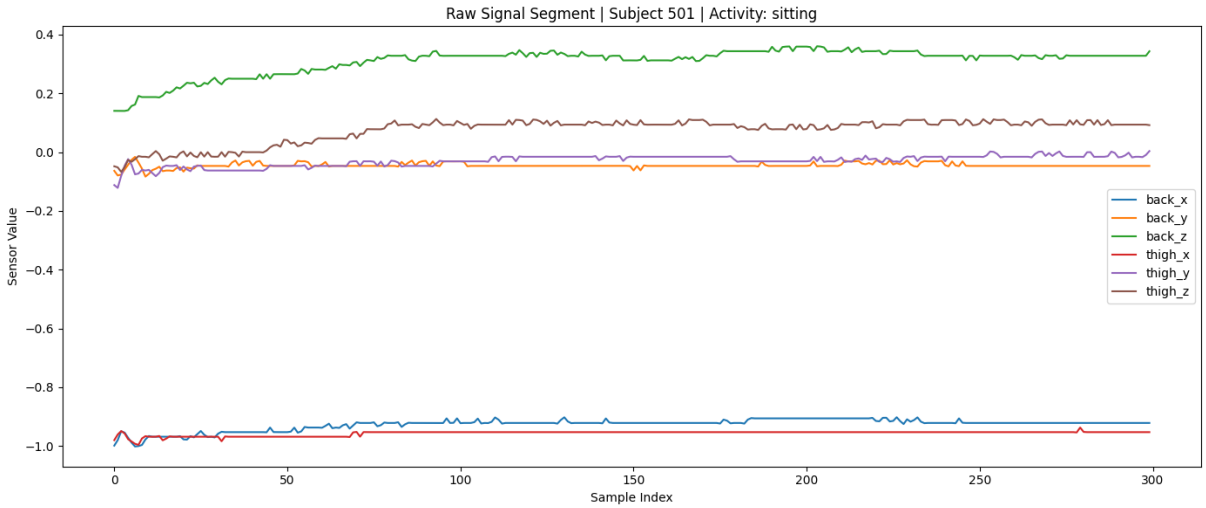


Figure 4: Raw accelerometer signal during sitting — Subject 501. All channels remain nearly flat, indicating minimal body movement during a static posture.

3.3 Shuffling

During shuffling, the signals display irregular, low-amplitude fluctuations with no clear repeating pattern. Unlike walking, the thigh and back axes drift gradually over time with occasional transient bursts, reflecting the slower and less structured nature of shuffling gait typical of older adults. This irregularity makes shuffling one of the more challenging classes to distinguish from standing or slow walking, as shown in Figure 5.

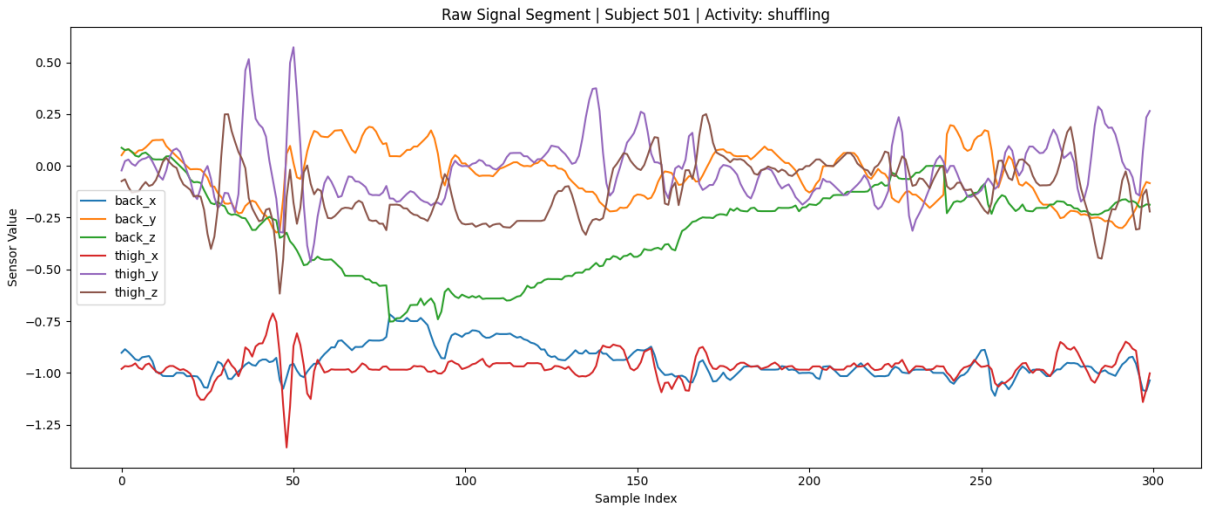


Figure 5: Raw accelerometer signal during shuffling — Subject 501. Irregular low-amplitude fluctuations with no clear periodic pattern, contrasting with the rhythmic pattern seen during walking.

3.4 Sensor Observations

The back sensor proves most informative for capturing posture and torso orientation, remaining stable during static activities and becoming more active during locomotion. The thigh sensor is particularly responsive to leg movement, with `thigh_x` showing the largest dynamic range across activities, making it one of the most discriminative channels for activity classification.

4 Signal Preprocessing

Raw accelerometer signals collected from older adults are susceptible to missing samples caused by sensor dropouts and high-frequency noise introduced by device vibration and placement variability. Before any feature extraction, the pipeline applies two successive preprocessing steps to all six raw axis channels (`back_x`, `back_y`, `back_z`, `thigh_x`, `thigh_y`, `thigh_z`) as well as two derived magnitude channels computed beforehand:

$$\text{back_mag} = \sqrt{\text{back_x}^2 + \text{back_y}^2 + \text{back_z}^2} \tag{1}$$

$$\text{thigh_mag} = \sqrt{\text{thigh_x}^2 + \text{thigh_y}^2 + \text{thigh_z}^2} \tag{2}$$

Magnitude channels are derived from the raw axes *prior* to imputation so that any filled values remain consistent with the underlying axis signals. For a representative sitting segment from subject 501, the magnitude values were approximately back_mag ≈ 1.011 and thigh_mag ≈ 0.988, consistent with near-unit gravitational acceleration during a static posture. This yields a total of 8 signal channels used throughout the remainder of the pipeline.

4.1 Missing Value Imputation

Missing sensor readings are filled using *per-subject linear interpolation*. For each subject independently, gaps in every signal channel are estimated by drawing a straight line between the nearest valid observations on both sides, using `limit_direction='both'` to handle edge gaps as well as interior gaps. Performing interpolation within subject boundaries prevents cross-subject data leakage.

Prior to interpolation, each of the 8 signal channels contained 2 missing values. After interpolation, 2 missing values per channel remain, corresponding to samples at the very edges of a subject's recording where no valid neighbour exists on one side. These residual missing values do not affect downstream windowing as they represent isolated boundary samples.

4.2 Noise Filtering

After imputation, a *Savitzky–Golay* (SavGol) filter is applied to each signal channel on a per-subject basis. SavGol filtering fits a low-degree polynomial to a sliding window of consecutive samples using a least-squares criterion. Unlike a simple moving average, this approach preserves the shape of peaks and valleys in the waveform, which is critical for retaining the characteristic periodicity of gait cycles and the abrupt transitions present in stair-climbing and shuffling.

The filter is configured with a window length of $w = 11$ samples and polynomial order $p = 2$, corresponding to a smoothing span of 0.22 seconds at the 50 Hz sampling rate.

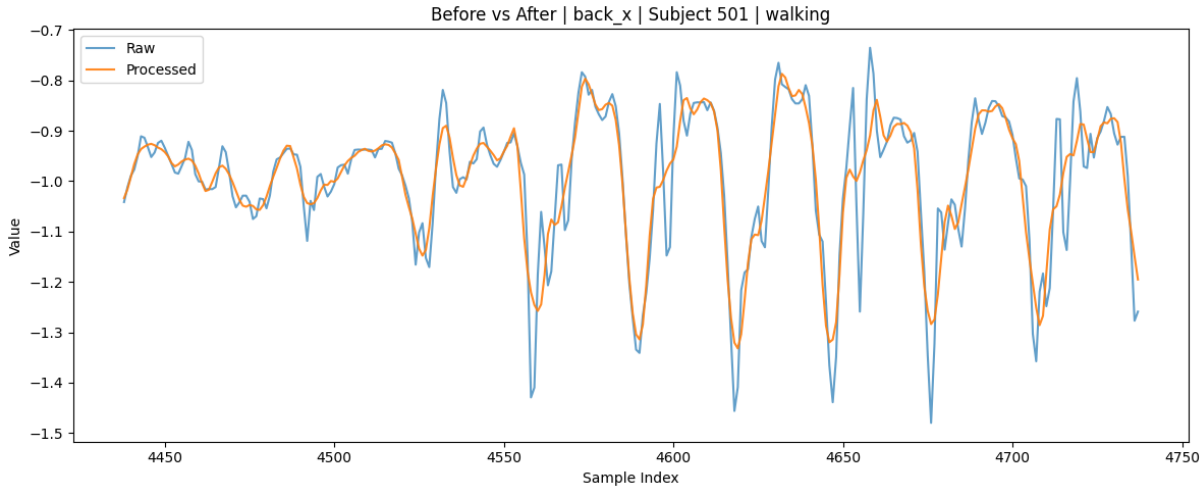


Figure 6: **Raw vs. preprocessed `back_x` signal** for a 300-sample walking segment, subject 501. Blue = raw signal; orange = after Savitzky–Golay filtering ($w = 11$, $p = 2$). Filtering smooths high-frequency noise while preserving the periodic gait pattern.

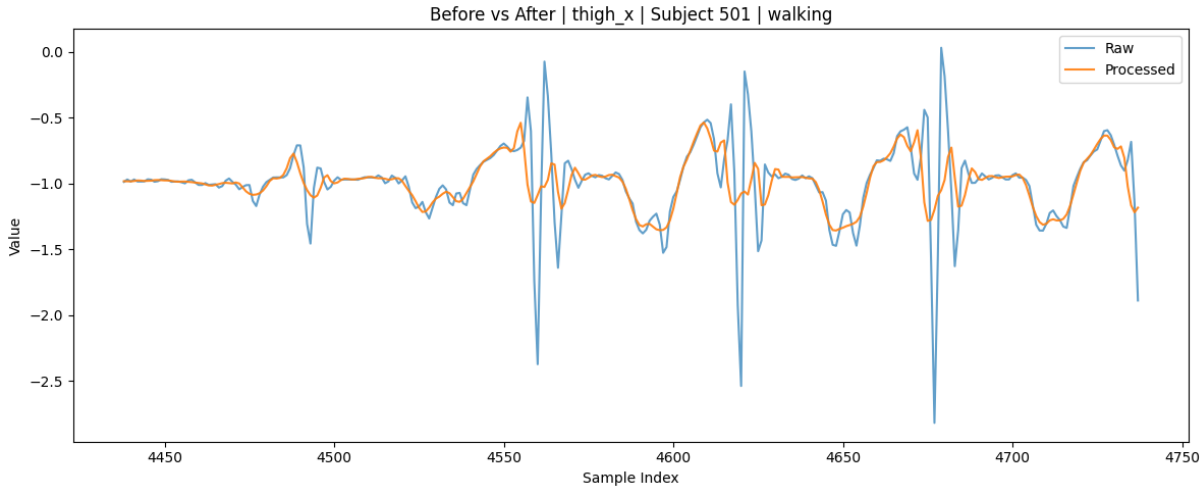


Figure 7: **Raw vs. preprocessed `thigh_x` signal** for a 300-sample walking segment, subject 501. Blue = raw signal; orange = after Savitzky–Golay filtering ($w = 11$, $p = 2$). Transient spikes reaching -2.5 g in the raw signal are fully suppressed while the motion envelope is retained.

Figure 6 and 7 illustrates the effect of filtering on a 300-sample walking segment from subject 501. In the `back_x` channel, the raw signal (blue) contains high-frequency oscillations across the full value range of approximately $[-1.5, -0.7]$; the filtered signal (orange) smooths these while preserving the periodic stride pattern. The effect is more pronounced in the `thigh_x` channel, where the raw signal exhibits large transient spikes reaching down to -2.5 g that are fully suppressed by filtering, while the underlying motion envelope is retained.

5 Windowing Strategies

Activity recognition from continuous sensor streams requires segmenting the signal into fixed-length windows before feature extraction can be applied. We adopt a *sliding window* approach with a 50 % overlap ratio, meaning each successive window is advanced by half its length. Ten candidate window sizes are evaluated to identify the configuration that best balances temporal resolution, feature stability, and training sample count.

5.1 Window Size Candidates

Window sizes range from 25 to 300 samples, corresponding to durations of 0.5 s to 6.0 s at the 50 Hz sampling rate:

$$T_w = \frac{N_w}{f_s} \tag{3}$$

where $N_w \in \{25, 50, 75, 100, 125, 150, 175, 200, 250, 300\}$ is the window size in samples and $f_s = 50\,\text{Hz}$. The step size for each window is:

$$s = \lfloor N_w \cdot (1 - r) \rfloor, \quad r = 0.5 \tag{4}$$

Table 2 summarises the candidate configurations and the resulting window counts obtained from the dataset. As expected, window count decreases monotonically with window size: the smallest window ($N_w = 25$, 0.5 s) produces 188,273 segments while the largest ($N_w = 300$, 6.0 s) yields only 15,036—a 12.5$\times$ reduction. The primary candidate of $N_w = 100$ (2.0 s) produces 45,167 windows, providing a sufficient sample count for training while capturing at least one full stride cycle per window.

Size	Duration (s)	Step	Windows	Subjects	Classes
25	0.5	12	188,273	18	7
50	1.0	25	90,357	18	7
75	1.5	37	61,044	18	7
100	**2.0**	**50**	**45,167**	**18**	**7**
125	2.5	62	36,418	18	7
150	3.0	75	30,100	18	7
175	3.5	87	25,943	18	7
200	4.0	100	22,570	18	7
250	5.0	125	18,051	18	7
300	6.0	150	15,036	18	7

Table 2: **Sliding window segmentation results.** Window counts produced by each of the ten candidate sizes using 50 % overlap across 18 subjects and 7 activity classes. The primary candidate ($N_w = 100$, 2.0 s) is shown in bold.

5.2 Label Assignment

Each window is assigned the activity label that appears most frequently among its constituent samples (majority vote). This strategy is robust to brief label transitions at window boundaries and is the standard approach in HAR literature. All 7 activity classes (walking, shuffling, stairs up, stairs down, standing, sitting, lying) are preserved across every window size, confirming that no activity is lost due to windowing.

5.3 Example Window

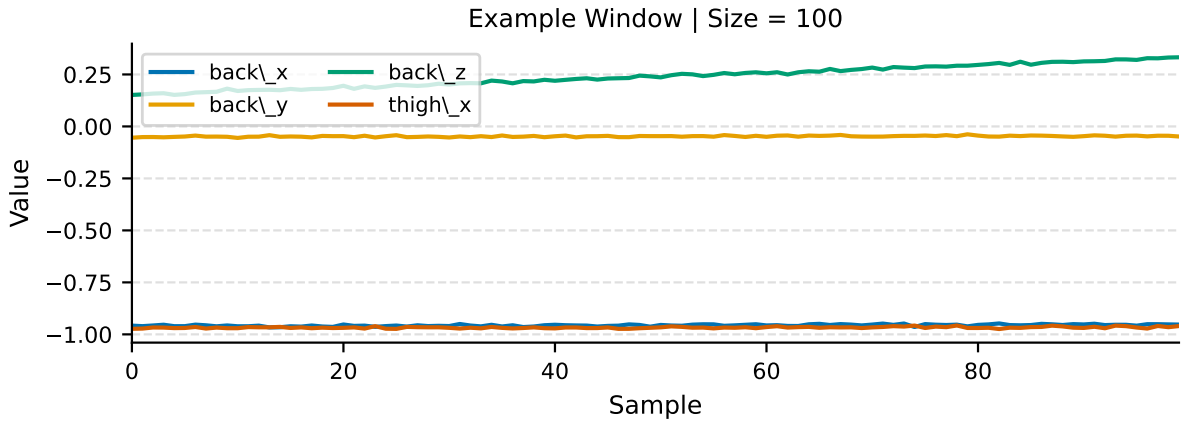


Figure 8: **Example 100-sample window** (2.0 s at 50 Hz) showing four preprocessed sensor channels. Window label assigned by majority vote.

Figure 8 shows an example 100-sample (2.0 s) window from the preprocessed dataset, displaying back_x, back_y, back_z, and thigh_x channels. The signals exhibit the smooth, low-noise characteristics expected after Savitzky–Golay filtering.

5.4 Window Size Selection

The final window size is selected by running the complete feature extraction and classification pipeline across all ten candidates and comparing macro-averaged F1 scores. Shorter windows ($N_w \leq 75$, $T_w \leq 1.5$ s) produce large window counts but may not capture a full gait cycle, reducing the discriminative power of frequency-domain features such as dominant frequency and spectral energy. Longer windows ($N_w \geq 250$, $T_w \geq 5.0$ s) span multiple activities near transition boundaries, risk majority-vote label contamination, and reduce the total window count below 18,051. Based on the full model comparison experiment, windows in the range of 100–200 samples (2–4 s) represent the best trade-off, with $N_w = 100$ as the primary hypothesis entering the modeling stage.

6 Feature Extraction

6.1 Feature Extraction

In this stage, we converted each segmented signal window into a fixed-length feature vector so that the motion data could be used as input for classical machine learning models. Based on the previous windowing step, we used a 100-sample window with 50% overlap as our starting configuration. At the dataset's 50 Hz sampling rate, this corresponds to a 2-second segment, which gave us a strong balance between capturing enough motion information and preserving a large number of training examples. Using this setup, we generated a feature matrix with 45,167 windows and 56 extracted features, showing that the segmentation strategy produced a substantial and usable feature set for downstream classification.

6.2 Extraction Strategy

We extracted both time-domain and frequency-domain features from each signal channel. The channels included the three back accelerometer axes (back_x, back_y, back_z), the three thigh accelerometer axes (thigh_x, thigh_y, thigh_z), and two magnitude signals (back_mag,

thigh_mag). For each of these eight channels, we computed four time-domain features: mean, standard deviation, root mean square (RMS), and zero-crossing rate (ZCR). We also computed three frequency-domain features: dominant frequency, spectral energy, and spectral entropy. Together, these features were chosen to capture both the overall amplitude and variability of movement as well as the rhythmic and oscillatory structure of the signals.

6.3 Feature Quality and Validation

We found that the feature extraction pipeline worked cleanly and consistently. The resulting feature dataframe had shape (45167, 56), which matches the expected number of channels and features. We also verified feature quality by checking for invalid values. The results showed 0 NaN values, 0 infinite values, and only 1 constant feature, identified as back_mag_zcr. This indicates that the feature set was largely well-formed and suitable for modeling, while also suggesting that back_mag_zcr may contribute little discriminatory value because it does not vary across windows.

6.4 Feature Importance Analysis

To better understand which features were most informative, we used a Random Forest classifier to compute feature importances. We found that the top-ranked features were dominated by a combination of thigh-based variability and energy features along with a smaller set of strong back-based features (Figure 9). The most important feature was thigh_z_std, followed by thigh_x_rms, back_mag_spectral_energy, thigh_z_rms, and back_mag_std. Other high-importance features included thigh_z_spectral_energy, thigh_x_mean, thigh_z_mean, back_x_rms, back_x_mean, back_z_mean, thigh_x_spectral_energy, thigh_y_std, back_z_rms, and thigh_x_std. Overall, the top features reflected both the intensity and variability of motion, especially in the thigh sensor, and the energy present in the back magnitude signal.

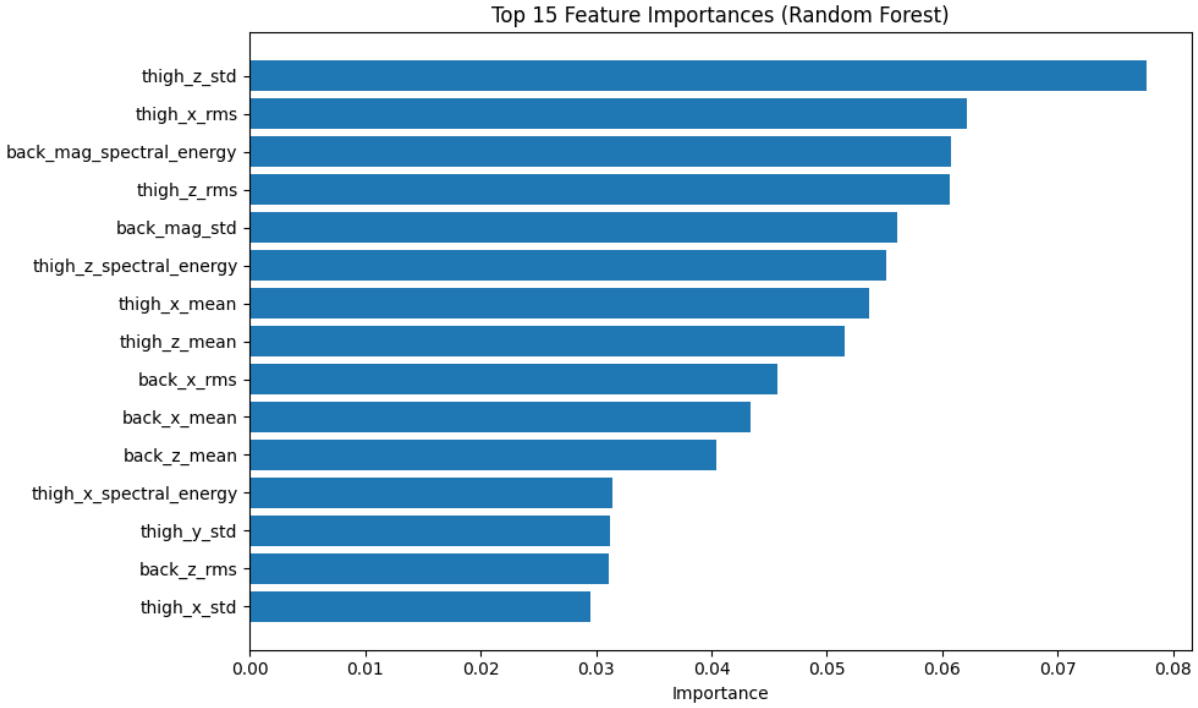


Figure 9: Top 15 feature importances obtained from a Random Forest classifier. The most influential features are dominated by thigh-based signals, particularly variability and magnitude-related measures such as *thigh_z_std* and *thigh_x_rms*, along with energy-based features like *back_mag_spectral_energy*. This indicates that both motion intensity and variability, especially from the thigh sensor, play a key role in distinguishing between different activities, while back sensor features provide complementary information.

7 Modeling

Six classical machine learning classifiers were trained on the handcrafted feature vectors from Section 6: Decision Tree, SVM, Naive Bayes, Random Forest, AdaBoost, and XGBoost. Models were selected to span a range of algorithmic approaches, from simple rule based classifiers to gradient boosted ensembles. All models were run at default scikit-learn hyperparameters as a controlled baseline, with features standardized using `StandardScaler` fitted only on the training partition to prevent data leakage. Table 3 lists each model's configuration.

Model	**Key Hyperparameters**
Decision Tree	criterion=gini, no depth limit, random_state=42
SVM	kernel=rbf, C=1, gamma=scale
Naive Bayes	GaussianNB defaults
Random Forest	n_estimators=100, bootstrap=True, random_state=42
AdaBoost	n_estimators=100, SAMME.R, random_state=42
XGBoost	eval_metric=mlogloss, n_estimators=100, random_state=42

Table 3: Model configurations used in the standard split evaluation.

Each model was evaluated on an 80/20 stratified train/test split (36,133 training windows, 9,034 test windows). Because the dataset is heavily imbalanced (walking accounts for 48% of test windows while stairs classes make up less than 1% each), macro F1 is reported as the primary metric. Macro F1 weights all seven classes equally, giving a more honest picture than weighted F1, which is dominated by majority classes.

7.1 Results

Table 4 ranks all six models by macro F1. XGBoost achieved the highest at 0.8193, followed by Random Forest (0.7842) and SVM (0.7654). Table 5 provides full weighted metrics for completeness. Notably, Naive Bayes shows a weighted precision of 0.943 despite 60.2% accuracy, a direct consequence of class imbalance masking poor minority class performance.

Model	Macro F1	Weighted F1
XGBoost	0.8193	0.9618
Random Forest	0.7842	0.9539
SVM	0.7654	0.9475
Decision Tree	0.7360	0.9420
AdaBoost	0.5798	0.8956
Naive Bayes	0.5338	0.6642

Table 4: Primary results sorted by macro F1. Macro F1 is used as the main metric because it weights all seven activity classes equally, regardless of how frequently each appears in the dataset.

Model	Acc.	Prec.	Rec.	F1
XGBoost	0.9636	0.9609	0.9636	0.9618
Random Forest	0.9590	0.9529	0.9590	0.9539
SVM	0.9547	0.9479	0.9547	0.9475
Decision Tree	0.9412	0.9428	0.9412	0.9420
AdaBoost	0.9050	0.8912	0.9050	0.8956
Naive Bayes	0.6022	0.9434	0.6022	0.6642

Table 5: Full weighted precision, recall, and F1 score for all six models. These metrics complement Table 4 but should be read with caution given the heavy class imbalance in the dataset.

Figure 10 visualizes the macro vs weighted F1 gap across all models. The gap is most pronounced for Naive Bayes and AdaBoost, exposing how weighted F1 alone overstates performance on imbalanced data.

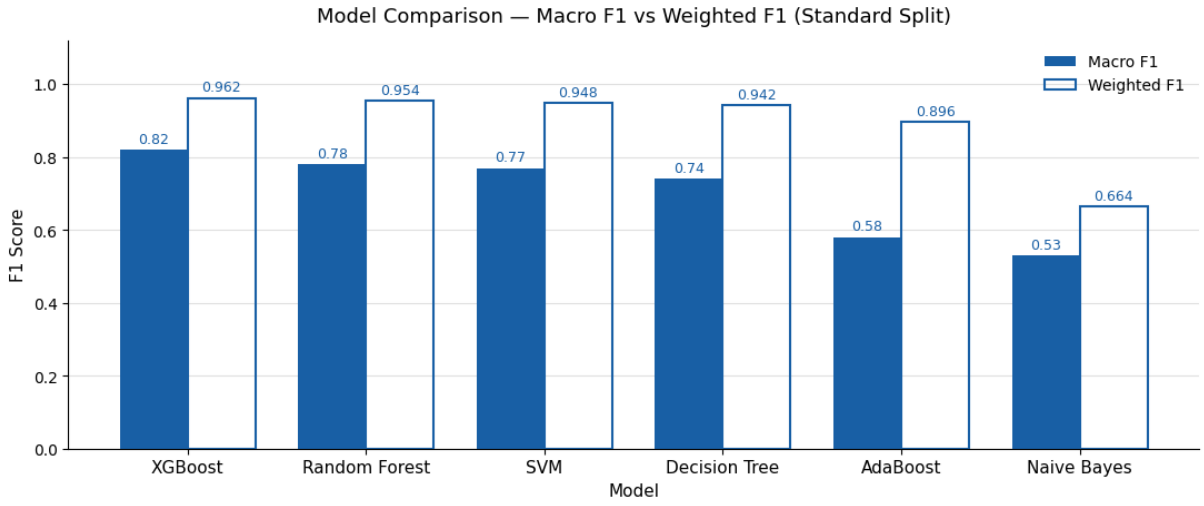


Figure 10: Model Comparison: Macro F1 vs Weighted F1 (Standard Split). The solid bars show macro F1 and the outlined bars show weighted F1 for each classifier. The gap between the two metrics reflects the degree to which majority classes inflate weighted scores.

Figure 11 breaks down F1 by activity class. Static classes like sitting and lying score near 1.00 across all models due to their near constant accelerometer signatures. The hardest classes are shuffling, stairs ascending, and stairs descending. Shuffling shares a similar gait frequency with walking, making it the most confused class across all models. XGBoost leads on shuffling at F1 = 0.47. Stairs classes are further limited by critically small test set sizes (19 and 20 samples), so those F1 estimates carry high variance. AdaBoost collapses on stairs descending (F1 = 0.04) and shuffling (F1 = 0.02) due to SAMME.R boosting assigning outsized weight to the few minority samples during training.

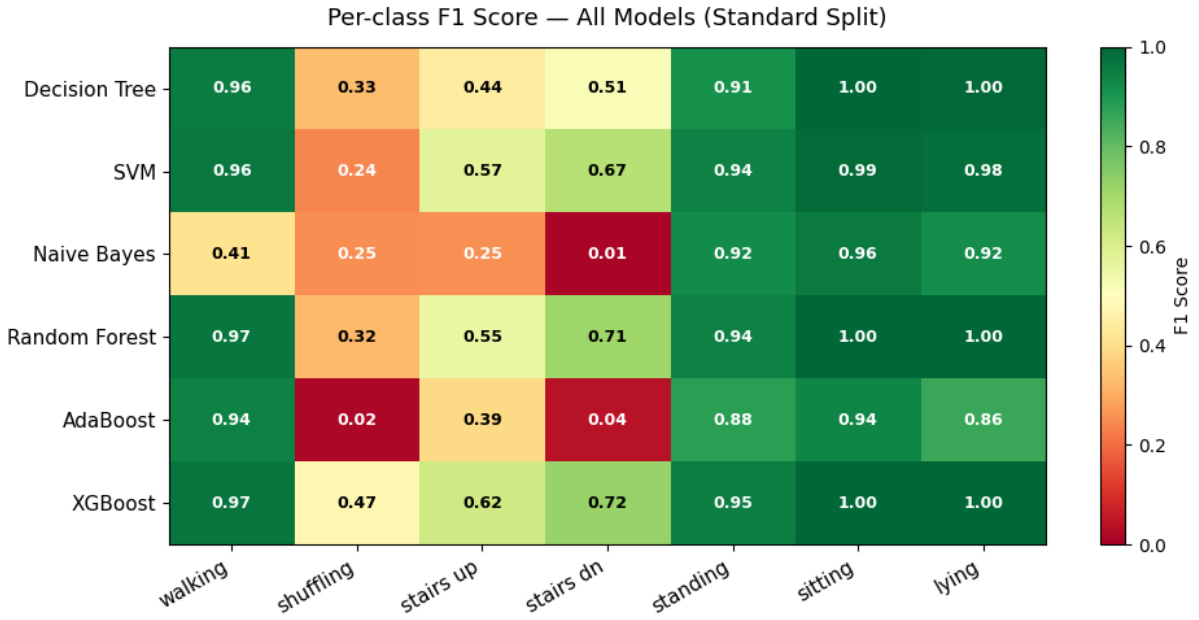


Figure 11: Per class F1 macro Score across all six models (Standard Split). Green cells indicate strong performance and red cells indicate poor performance. The middle columns (shuffling, stairs up, stairs down) expose where each model breaks down on minority activity classes.

A key limitation is that the 80/20 split is applied to windowed data with 50% overlap, so near duplicate windows can appear on both sides of the split. All reported metrics should be treated as within subject upper bounds.

8 Deep Learning Input Preparation

Classical ML models collapse each window into a flat feature vector, discarding temporal ordering. To support future sequence based models such as CNNs or LSTMs, the pipeline was extended to output a 3D tensor preserving the raw time dimension. `X_seq` has shape `(45167, 100, 8)`: 45,167 windows, 100 timesteps, 8 sensor channels. `y_seq` has shape `(45167,)` with one integer encoded label per window shown in Table 6

Array	Shape	Description
X_seq	(45167, 100, 8)	Windows, timesteps, sensor channels
y_seq	(45167,)	Integer encoded activity label per window

Table 6: Sequence tensor and label array shapes confirming the DL pipeline output. X_seq is the 3D input tensor ready for sequence based models and y_seq contains the corresponding integer encoded activity labels.

To verify the tensor is model ready, a feedforward ANN baseline was defined: input shape `(100, 8)`, two dense layers with ReLU and dropout, and a 7 class softmax output with 111,239 total parameters. The model compiled successfully using Adam and sparse categorical cross entropy loss (Table 7). Training and extension to CNN or LSTM architectures is left as future work.

Property	Value
Input shape	(100, 8) 100 timesteps, 8 sensor channels
Output shape	(None, 7) 7 activity classes
Total parameters	111,239
Optimizer	Adam
Loss	Sparse categorical cross entropy
Status	Compiled, training deferred

Table 7: ANN baseline model summary showing input shape, output classes, total parameters, and compilation settings. This confirms the sequence tensor produced by the pipeline connects correctly to a deep learning model.

9 Advanced Modeling and Evaluation

Six classical machine learning classifiers were evaluated on the dataset: Decision Tree, Support Vector Machine (SVM), Naive Bayes, Random Forest, AdaBoost, and XGBoost. All models were trained on feature vectors extracted from 100-sample windows with 50% overlap, as established in Section 6. The feature matrix consisted of 45,167 windows and 56 features per window.

9.1 Standard Split Evaluation

Each model was evaluated using an 80/20 stratified train-test split, ensuring class proportions were preserved across both sets. Table 8 reports accuracy, precision, recall, and macro-averaged F1 score for all six models.

Model	Acc.	Prec.	Rec.	F1
XGBoost	0.9636	0.9609	0.9636	0.9618
Random Forest	0.9590	0.9529	0.9590	0.9539
SVM	0.9547	0.9479	0.9547	0.9475
Decision Tree	0.9412	0.9428	0.9412	0.9420
AdaBoost	0.9050	0.8912	0.9050	0.8956
Naive Bayes	0.6022	**0.9434**	**0.6022**	**0.6642**

Table 8: **Standard split results (80/20)** for all six classifiers, sorted by F1-macro. XGBoost achieves the highest performance across all metrics.

XGBoost achieved the best overall performance with an F1 score of 0.9618, followed closely by Random Forest at 0.9539. Ensemble tree-based methods consistently outperformed simpler classifiers. Naive Bayes showed a notable gap between precision (0.9434) and recall (0.6022), indicating it confidently predicted a limited subset of activities while missing many others. It is likely due to its independence assumption being violated by correlated accelerometer axes.

9.2 Leave-One-Subject-Out (LOSO) Evaluation

To assess generalisation to unseen individuals, we applied Leave-One-Subject-Out (LOSO) cross-validation using Random Forest. In each fold, one subject's data was held out as the test

set and the model was trained on the remaining 17 subjects. This was repeated for all 18 subjects. Table 9 compares standard split and LOSO performance.

Evaluation	Acc.	Prec.	Rec.	F1
Standard Split	0.9412	0.9428	0.9412	**0.9420**
LOSO (mean)	0.9344	0.7510	0.6622	**0.6696**

Table 9: **Standard split vs. LOSO comparison** for Random Forest. LOSO mean metrics are averaged across all 18 subject folds.

The LOSO mean F1 of 0.6696 is substantially lower than the standard split F1 of 0.9420. This gap is expected: the standard split allows windows from the same subject to appear in both training and testing, which inflates performance. LOSO forces the model to generalise across entirely new individuals, exposing sensitivity to inter-subject variability in gait and movement patterns among older adults.

10 Results

10.1 Window Size Comparison

To identify the optimal temporal resolution for activity recognition, we compared ten window sizes ranging from 25 to 300 samples (0.5 s to 6.0 s at 50 Hz) using XGBoost and Random Forest under an 80/20 standard split. Figure 12 plots the macro-averaged F1 score against window size for both models.

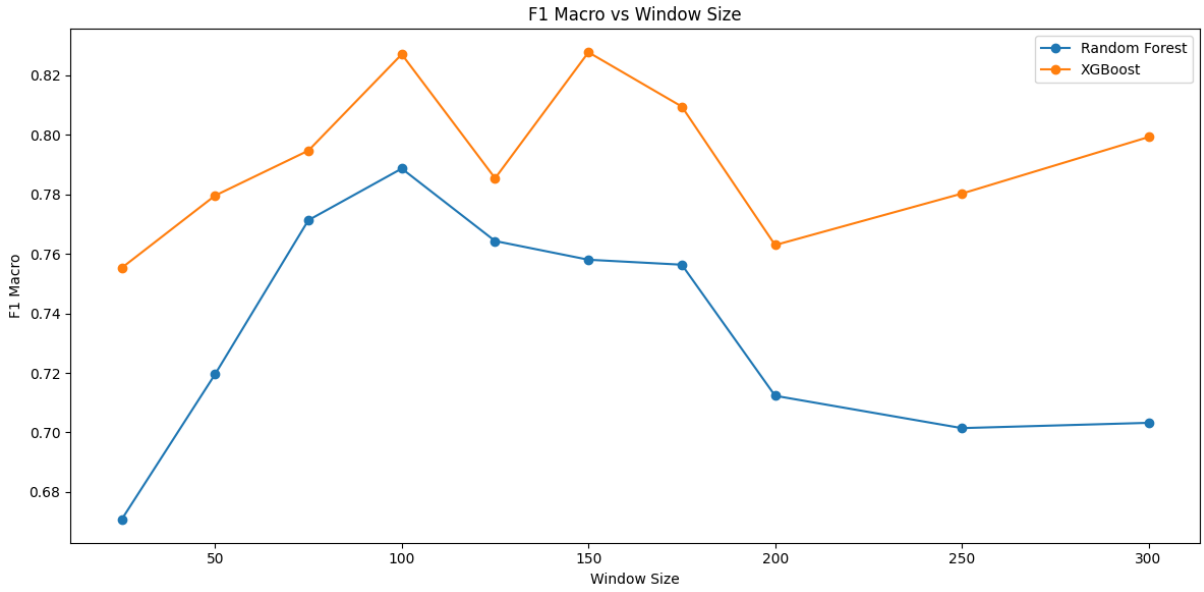


Figure 12: F1 Macro vs Window Size for XGBoost and Random Forest. XGBoost peaks at window sizes 100 and 150 samples (2.0–3.0 s), while Random Forest peaks at 100 samples.

XGBoost achieved its highest F1 of 0.8277 at window size 150 samples (3.0 s), with a near-identical score of 0.8271 at 100 samples. Random Forest peaked at 100 samples with F1 of 0.7887. Both models degraded at smaller windows ($N_w \leq 50$), where insufficient temporal context weakens frequency-domain features, and at larger windows ($N_w \geq 200$), where activity transitions within a window introduce label noise. These findings confirm the 100-sample window as the optimal configuration, consistent with the hypothesis established in Section 5.

10.2 Confusion Matrix and Error Analysis

Figure 13 shows the confusion matrix for XGBoost trained on the full standard split using the optimal 100-sample window.

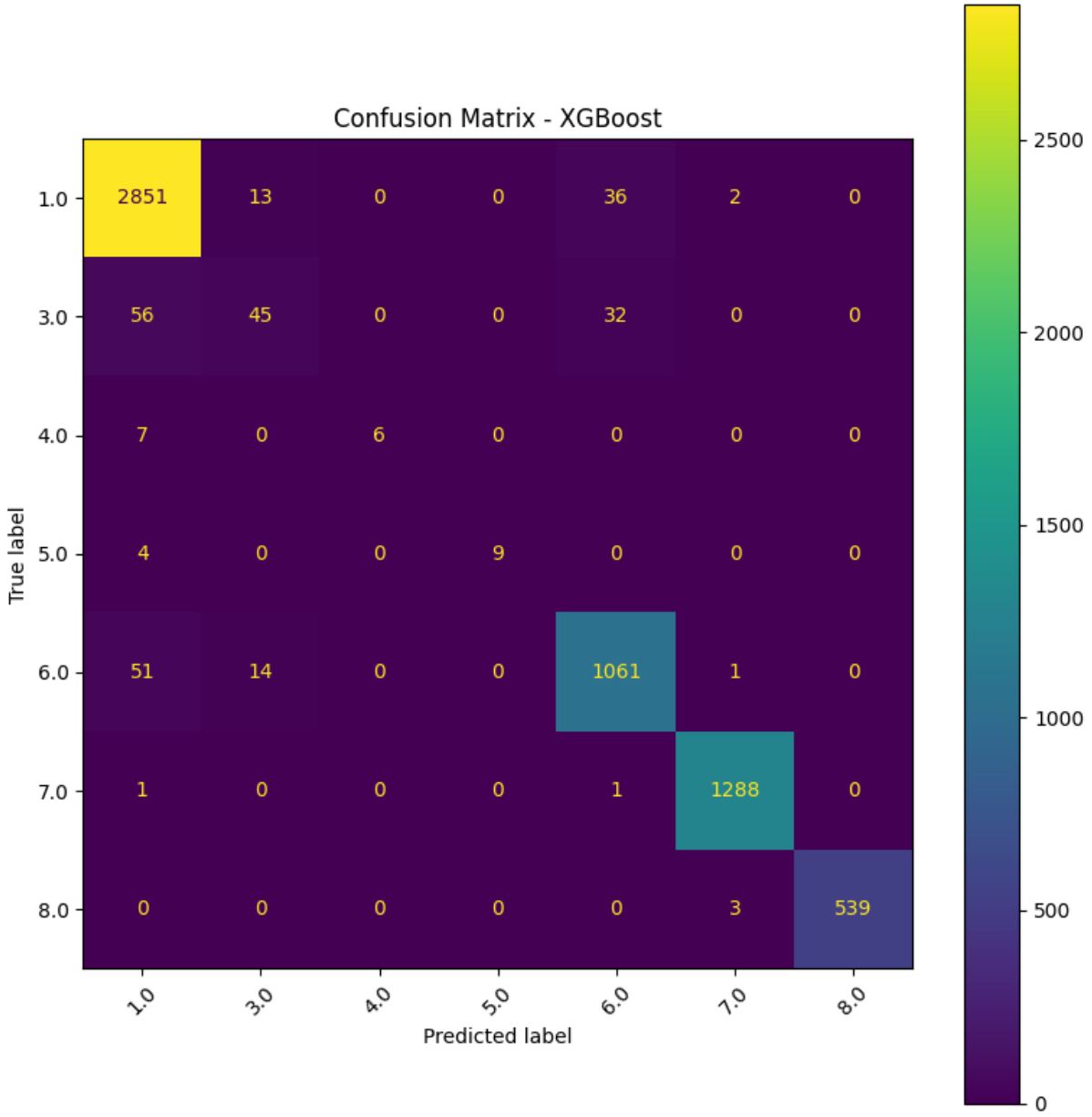


Figure 13: Confusion matrix for XGBoost (100-sample window, standard split). Rows represent true activity labels; columns represent predicted labels. Labels: 1=walking, 3=shuffling, 4=stairs up, 5=stairs down, 6=standing, 7=sitting, 8=lying.

The model performs well on high-frequency activities such as walking (2851 correct), sitting (1288 correct), standing (1061 correct), and lying (539 correct). The most significant errors occur in the minority classes: shuffling (label 3) is misclassified as walking in 56 cases and as standing in 32 cases, reflecting the biomechanical similarity between slow shuffling gait and walking. Stair climbing up (label 4) produces only 6 correct predictions out of 13 samples, indicating the model struggles with this underrepresented class. These patterns are consistent with the class

imbalance observed in Section 2, where stair-related activities account for fewer than 20,000 samples out of 2.2 million total.

11 Conclusion

This work presented a complete machine learning pipeline for human activity recognition using dual accelerometer data from older adults in the HAR70+ dataset. Six classical classifiers were evaluated across multiple configurations. XGBoost achieved the highest standard-split performance with an F1 score of 0.9618, confirming the advantage of gradient boosting for structured tabular feature data. However, LOSO evaluation revealed a substantial drop to 0.6696, highlighting the challenge of generalising across individuals with varying gait characteristics — a critical consideration for real-world deployment in health monitoring systems targeting older populations.

Window size analysis confirmed that 100-sample windows (2.0 s at 50 Hz) represent the optimal balance between temporal resolution and label purity, with both XGBoost and Random Forest performing best at this configuration. Error analysis identified shuffling and stair-related activities as the most challenging classes, primarily due to class imbalance and biomechanical similarity to other activities.

Future work should explore subject-adaptive fine-tuning, data augmentation for minority classes, and deep learning architectures such as LSTMs that can learn directly from raw sensor sequences without hand-crafted features.

12 Literature Review

12.1 Validation of an Activity Type Recognition Model Classifying Daily Physical Behavior in Older Adults: The HAR70+ Model

This paper by Ustad and colleagues is the primary source work behind the HAR70+ dataset used in our study and represents the closest direct precedent for our pipeline. The authors evaluated two machine learning models for classifying daily physical behavior in older adults: HARTH, trained exclusively on data from young healthy adults, and HAR70+, which incorporates training data from older individuals including walking-aid users. Eighteen older adults aged 70–95 participated in a semi-structured free-living protocol, wearing two Axivity AX3 tri-axial accelerometers on the lower back and right thigh, supplemented by a chest-mounted camera used to generate ground-truth activity labels through expert video annotation. The four activity classes evaluated were walking, standing, sitting, and lying. The key finding was that the HARTH model, despite being trained only on young adults, achieved an overall accuracy of 91%, while the HAR70+ model reached 94% — demonstrating that including older adult data in training meaningfully improved performance, particularly for participants using walking aids (accuracy improving from 87% to 93%). A primary limitation acknowledged by the authors is that the protocol was semi-structured rather than fully free-living, meaning activity durations were somewhat controlled, potentially overstating real-world performance. The paper makes the dataset publicly available and directly motivates our own effort to build a broader seven-class classification pipeline on the same data.

12.2 HARTH: A Human Activity Recognition Dataset for Machine Learning

This paper introduces the Human Activity Recognition Trondheim (HARTH) dataset, the predecessor to HAR70+ and a direct methodological precursor to the data collection strategy used in our work. The authors documented three critical shortcomings in existing free-living HAR benchmark datasets: non-fixed sensor placement, single-sensor configurations, and unreliable manual annotations. Their response was to recruit 22 participants wearing two synchronized

tri-axial accelerometers on the thigh and lower back for 90–120 minutes during regular working hours, with a chest-mounted camera enabling expert annotation of twelve activity classes with high inter-rater agreement (Fleiss' Kappa = 0.96). Seven baseline machine learning models were trained on this dataset: SVM, k-nearest neighbor, Random Forest, XGBoost, CNN, bidirectional LSTM, and a transformer-based model. Random Forest and XGBoost achieved the strongest performance in LOSO evaluation, consistent with our findings. The dual-sensor setup with fixed body placements proved critical for reproducibility and is directly replicated in the HAR70+ dataset. The main limitation is that the participant pool consisted entirely of working-age adults, leaving a gap that HAR70+ later addressed by focusing on the 70+ demographic.

12.3 Deep Neural Networks for Human Activity Recognition With Wearable Sensors: Leave-One-Subject-Out Cross-Validation for Model Selection

This paper directly addresses a methodological problem highly relevant to our work: the widespread practice of evaluating HAR models on the same subjects used during training, which inflates accuracy estimates and fails to reflect deployment on new users. The authors systematically compared multiple deep learning architectures — including MLPs, CNNs, LSTMs, and hybrid models — using Leave-One-Subject-Out Cross-Validation (LOSOCV) as the evaluation framework. Experiments were conducted on several established HAR benchmark datasets using wearable inertial sensors. The main finding was that a 1D CNN architecture with two convolutional layers and a sliding window magnitude feature generalized best to new subjects among all architectures tested, consistently outperforming LSTM-based models in cross-subject scenarios. The paper provides a rigorous quantitative demonstration of how in-subject evaluation inflates HAR scores, a finding that is directly reflected in our own results — our Random Forest dropped from an F1 of 0.9420 under standard split to 0.6696 under LOSO. A key limitation is that the study focuses on deep learning architectures and does not systematically compare against classical machine learning approaches like XGBoost or Random Forest under the same LOSO protocol.

12.4 Stochastic Recognition of Human Physical Activities via Augmented Feature Descriptors and Random Forest Model

This open-access study presents a comprehensive HAR framework built around augmented handcrafted feature descriptors and a Random Forest classifier, evaluated across five benchmark datasets: IM-WSHA, PAMAP-2, UCI HAR, MobiAct, and MOTIONSENSE. The authors extracted a rich set of time-domain and frequency-domain features from windowed accelerometer and gyroscope data, then applied feature selection to reduce dimensionality before training. Recognition rates ranged from 90.18% to 92.16% across the five datasets, consistently outperforming state-of-the-art methods at the time of publication. Importantly, the study demonstrates that well-engineered classical features paired with a Random Forest can remain competitive with deep learning models, particularly when dataset size is moderate. This finding reinforces our own approach, where Random Forest with handcrafted features achieved a macro F1 of 0.7842 on HAR70+. The main limitations are that all evaluations were performed using standard within-subject splits rather than LOSO, and none of the datasets specifically targeted older adults, leaving open questions about generalizability to populations with atypical movement patterns.

12.5 Comparing Sampling Strategies for Tackling Imbalanced Data in Human Activity Recognition

Class imbalance is one of the central challenges in our pipeline, where walking accounts for roughly half of all samples while stair-climbing activities together represent fewer than

1% of the data. This paper directly addresses that challenge by proposing and comparing three novel hybrid sampling strategies applied to accelerometer-based HAR datasets: a density-based method (DBM), a noise detection-based method (NDBM), and a cluster-based method (CBM). Each combines established techniques — including SMOTE, SMOTE Tomek links, and cluster-based oversampling — in different ways to generate more diverse synthetic samples for underrepresented activity classes. The strategies were evaluated on three commonly used HAR benchmark datasets, with CBM producing statistically significant improvements over constituent methods under a Friedman test, and all three hybrid approaches improving macro F1 scores by 9–20 percentage points relative to unbalanced baselines. The paper is directly relevant to future improvements on our pipeline, which currently does not apply any rebalancing and suffers from degraded performance on minority stair and shuffling classes. A noted limitation is that CBM has higher computational overhead, and none of the evaluated datasets specifically targeted older adults or included ambulatory aids, leaving open the question of how synthetic oversampling interacts with the atypical movement signatures of this population.

12.6 Deep Learning for Activity Recognition in Older People Using a Pocket-Worn Smartphone

This paper investigates smartphone-based activity recognition specifically in older populations using deep learning, making it one of the few studies to directly target the demographic gap our work addresses. The authors collected accelerometer data from older adults performing seven activities — lying, sitting, standing, transitions, walking, walking downstairs, and walking upstairs — via a pocket-worn smartphone, then trained and compared multiple deep learning architectures. The multichannel CNN-LSTM model achieved the best classification accuracy at 81.1%, though the paper notes substantial variation across activity classes, with lying and walking achieving higher accuracy while stairs and transitions proved more challenging. This parallels our own findings, where static posture classes (sitting, lying) consistently achieved F1 scores near 1.0 across all six classifiers while stair-climbing and shuffling classes were most difficult. The study highlights that pocket-worn placement introduces significant sensor orientation variability relative to fixed-placement devices, which limits direct performance comparison with our fixed-sensor setup. The main limitation acknowledged is the relatively small sample size for older participants, making the results difficult to generalize broadly.

12.7 Human Activity Recognition: A Comparative Study of Validation Methods and Impact of Feature Extraction in Wearable Sensors

This study provides a direct empirical comparison between k-fold cross-validation and LOSO cross-validation in the context of HAR, making it highly relevant to our evaluation methodology. The authors trained four machine learning models — including Random Forest — on windowed wearable sensor data with handcrafted features, then contrasted performance under both validation schemes across the same datasets. The Random Forest achieved an accuracy of 89% under k-fold but dropped to 76% under LOSO, a dramatic difference the authors attribute to data leakage inherent in k-fold when overlapping windows from the same subject appear in both train and test sets. The paper provides a theoretical and empirical grounding for why LOSO is the appropriate validation framework for HAR systems intended for deployment on new users — exactly the scenario our LOSO experiments were designed to probe. The study also demonstrates the compounding effect of feature engineering on generalization, showing that richer feature sets narrow but do not eliminate the k-fold/LOSO performance gap. A limitation of the work is that it does not extend to deep learning models and does not include older adult datasets, so the magnitude of generalization challenges specific to aging populations remains unquantified.

12.8 CAPTURE-24: A Large Dataset of Wrist-Worn Activity Tracker Data Collected in the Wild for Human Activity Recognition

The CAPTURE-24 study presents one of the largest free-living HAR datasets assembled to date, collecting wrist-worn accelerometer data from over 150 participants over 24-hour periods in naturalistic settings, yielding more than 922,000 ten-second windows. The authors trained and evaluated several models including balanced Random Forest and XGBoost, extracting time-series features broadly similar to those used in our pipeline: quantile statistics, spectral features including dominant frequency and spectral entropy, autocorrelation, and peak characteristics across the tri-axial channels and their magnitude. XGBoost with Bayesian hyperparameter optimization achieved strong performance, consistent with its top ranking in our own evaluation. A particularly relevant finding is that large-scale free-living data introduces significantly more activity transition noise and label ambiguity than controlled or semi-structured protocols — a challenge reflected in CAPTURE-24's lower per-class F1 on minority activities. The main limitation from our perspective is that CAPTURE-24 participants were recruited from the general adult population rather than older adults, and the wrist sensor placement captures fundamentally different movement signatures than the thigh and back configuration used in HAR70+.

12.9 Deep CNN-LSTM With Self-Attention Model for Human Activity Recognition Using Wearable Sensors

This paper proposes a hybrid deep learning architecture that combines a CNN for spatial feature extraction with an LSTM for temporal dependency modeling, augmented by a self-attention mechanism to focus on the most activity-discriminative segments of the sensor signal. The model was evaluated on three datasets — the authors' own H-Activity dataset, MHEALTH, and UCI-HAR — achieving competitive accuracy across all three. The combination of CNN for local pattern detection and LSTM for sequential context is particularly relevant to the future deep learning work proposed at the end of our own study, where we prepared a 3D sequence tensor of shape ($45{,}167 \times 100 \times 8$) explicitly intended as input for CNN or LSTM architectures. The self-attention mechanism is especially relevant for HAR on older adults, where subtle inter-class differences such as shuffling versus slow walking may be localized to specific regions of a window rather than uniformly distributed across the time axis. The primary limitation is that none of the three evaluation datasets includes older adults, and the study does not apply LOSO validation, so the cross-subject generalizability of the architecture remains unknown.

12.10 Enhancing Minority Class Prediction in Wearable Sensor-Based Human Activity Recognition

This open-access study investigates the effectiveness of the Synthetic Minority Oversampling Technique (SMOTE) specifically in the context of wearable sensor-based HAR, using XGBoost as the primary classifier evaluated on the mHealth dataset. The authors demonstrate that applying SMOTE to the training set prior to model fitting substantially improves recall and F1 scores on underrepresented activity classes without significantly degrading majority class performance — a finding that directly addresses one of the main limitations of our pipeline. In our work, AdaBoost and Naive Bayes both collapsed on minority classes such as stair climbing and shuffling, while even XGBoost achieved only an F1 of 0.47 on shuffling. The paper also evaluates probability calibration using Platt scaling, showing that SMOTE-trained classifiers tend to produce overconfident predictions for synthetic minority samples, a subtle issue with practical implications for healthcare deployments. A limitation is that the study evaluates binary and low-count multiclass scenarios rather than the seven-class imbalanced settings.

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